

Austin House Price Prediction Contest

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Our goal was to create a housing price prediction model using housing data of the Austin area, scraped from Zillow. To arrive at the final model, we performed data wrangling and feature engineering, then tested both regression and tree-based methods.

Data Preprocessing

To create the final working data set, we removed, modified, and added features:

- 1-hot encoded the zip codes, and removed those that occurred less than 20 times in the data set. These were zip codes 78734, 78742, 78652, 78719, and 78738.
- Created a list of positively and negatively associated words based on online articles¹⁻³ and our own ideas about Austin-specific keywords. We then parsed the descriptions of each listing to find occurrences of these words. We removed words from the list with less than 20 occurrences.
- We removed columns that couldn't be coded numerically or categorically (`description`, `address`, `saleDate`, `saleMonth`, `saleYear`). We also removed `homeType` because it only had one level.

Feature Engineering

- Constructed two new features using `saleDate` and `yearBuilt`: `yearsOld` and `days_since_sale`. For the sake of these two new features, we assumed the data had been collected in 2022.
- Created log features from these: `lotSizeSqFt`, `livingAreaSqFt`, `avgSchoolSize`.
- Added polynomial features from these: `yearsOld`, `avgSchoolRating`.
- Used geographical coordinates and information from the city of Austin⁴ to assign neighborhoods to house listings.

The majority of the listings were outside of the regions defined by the city of Austin. We tried alternatives for identifying neighborhoods using *osmnx* and using natural clusters from the spatial distribution from Figure 1, but were unable to classify most of the listings in a way that meaningfully affected our RMSE.

- Lastly, we created two different versions of the data set. One encoding each positive and negative word as a binary feature, and another that aggregated these into a positive word count and negative word count to be used as numerical features.

Modeling Approach

We tested a variety of models before selecting the final one:

- LASSO and Ridge → We started with these to get some insight about features we could eliminate, but this only removed a few neighborhoods (Hancock, North Loop, and West University to name a few) and zip codes. However, we couldn't find a strong reason to exclude these from the more complex model types.
- XGBoost → We tuned the hyperparameters across a range of values: number of trees (100 or 300), maximum tree depth (3, 6, or 10), learning rate (0.1 or 0.3), row subsampling (70% or 100%), and feature subsampling (70% or 100%). Despite trying so many combinations, XGBoost still underperformed compared to our other models.
- Random forest → Exhaustive parameter tuning required astronomical compute time (not worth the trade-off in marginally improving RMSE). So, we used heuristics, but these also weren't improving RMSE all that much, and were still subject to systematic errors made by our initial assumptions and feature design. Additionally, it still struggled with high prices (cone shaped residuals). With all of these shortcomings, RF still became our best-performing model.

Results and Evaluation

Our best-performing model was Random Forest using the aggregate NLP numeric features and the set of neighborhoods defined by the Austin city website.

- RMSE: \$160,189.33
- Even though this model performed best overall, it still left some open questions about how neighborhoods influenced predictions. Some neighborhoods showed a lot of variability, so it might have helped to explore other ways of grouping homes. Additionally, over 60% of the homes in our dataset didn't belong to any defined neighborhood, which limited how much we could rely on that feature.
- Using feature importance (measured by mean decrease in variance), the most important features were `livingAreaSqFt`, `log_livingAreaSqFt`, `avgSchoolRating`, `avgSchoolRating_sq`, and `latitude`. The transformed features are highly correlated, so it's unsurprising they appear together at the top. It is more interesting that latitude is an important feature, this suggests there is a general East/West trend in housing prices, something that checks out anecdotally, with East Austin being historically poorer.
- Lastly, our model continues to struggle with high-priced houses, with less accurate predictions and subsequently larger residuals, as seen in Figures 2 and 3.

Original Feature	Transformation	New Feature(s)
saleDate, yearBuilt	Arithmetic	yearsOld, days_since_sale
lotSizeSqFt, livingAreaSqFt, avgSchoolSize	Log	log_lotSizeSqFt, log_livingAreaSqFt, log_avgSchoolSize
yearsOld, avgSchoolRating	Polynomial	yearsOld_sq, avgSchoolRating_sq
description	Binary	has_greenbelt, has_potential, has_...
	Count aggregation	word_count_good, word_count_bad
latitude, longitude	Spatial mapping	neighborhood

Table 1: Summary of feature transformations.



Figure 1: Home listing locations and Austin neighborhoods boundaries

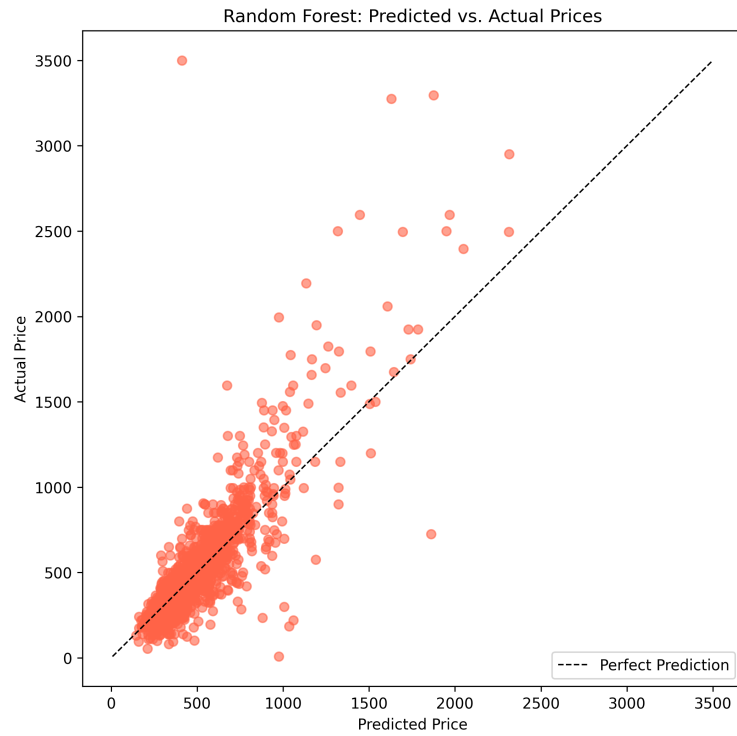


Figure 2: Predicted price vs. Actual price

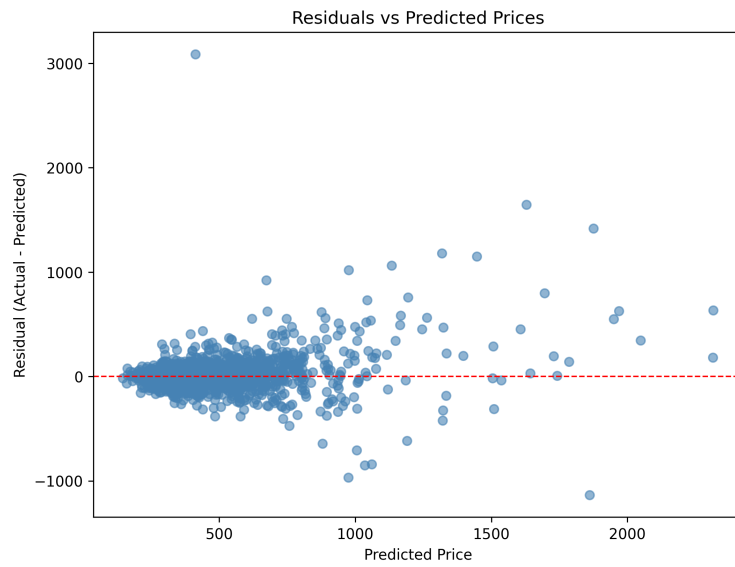


Figure 3: Residuals of RF model

Acknowledgments

We acknowledge the use of generative AI (ChatGPT) in the creation of this LaTeX template, tables, and significant portions of code, including data wrangling, neighborhood clustering, plotting, and model tuning.

References

1. Zillow. (2022). *15 real estate words that add value to your home listing*. <https://www.zillow.com/learn/15-words-that-add-value/>
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3. Hoffman, A. (2024). *12 red flags to watch for in real estate listings, according to agents*. House Beautiful. <https://www.housebeautiful.com/design-inspiration/real-estate/a64957249/real-estate-listing-warning-signs/>
4. City of Austin. (2025). *Boundaries: City of Austin Neighborhoods*. <https://data.austintexas.gov/Locations-and-Maps/Boundaries-City-of-Austin-Neighborhoods/inrm-c3ee>